Database Configuration with ODBC

Speaker Notes



INSIGHT
AND
OVERVIEW

List of topics

- What is ODBC? (slide)
- The IGSS ODBC Driver (slide)
- IGSS depends on the ODBC Server (slide)
- Configure a project from a database
- Create a new diagram and an object
- Insert the new objects in Definition
- The ODBC database structure

ODBC and the IGSS ODBC Driver

- ODBC (Open DataBase Connectivity)

  A Microsoft standard for communicating with different database products using SQL (Standard Query Language).

- IGSS ODBC Driver

  The driver makes IGSS data appear as a relational database.

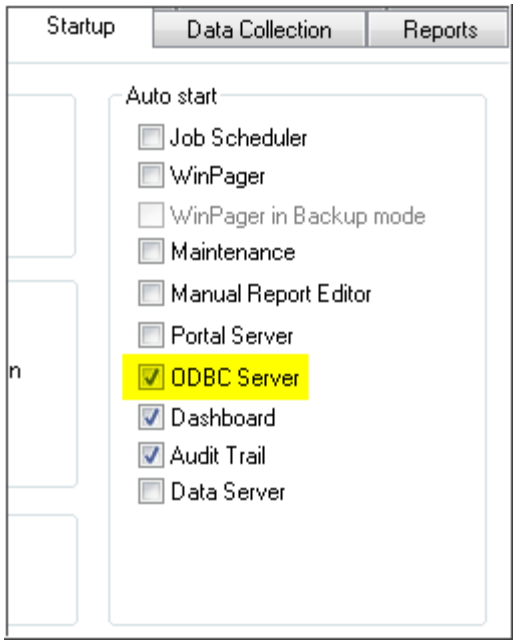
  The IGSS system designer can then create a new database and link to the current IGSS project.

  Any database product supporting SQL can be used (e.g. Microsoft Access, Microsoft Excel or Oracle)

IGSS depends on the ODBC Server

The IGSS ODBC Server must be running in order to use these features:

- IGSS Dashboard
- Object Historian in Supervise
- Embedded graphs
- Report Management System (RMS)
- Snapshot reports



IMPORTANT:
Make sure that ODBC Server is selected under Auto Start on the Startup tab in System Configuration.

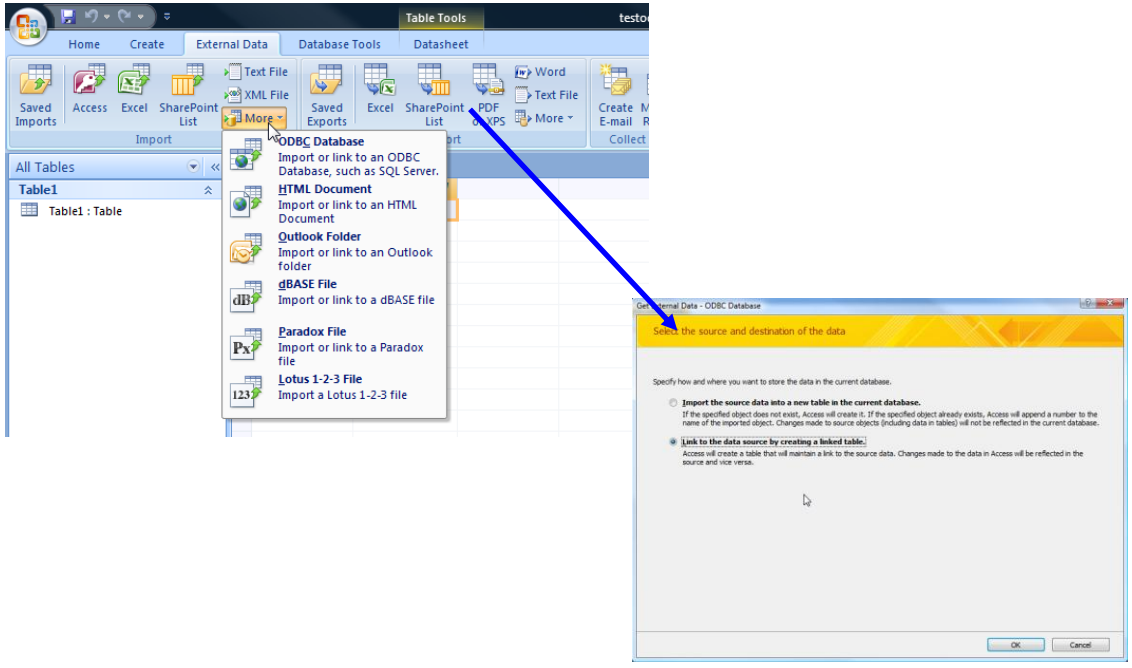
Configure a project with ODBC (1)

1. Close all IGSS programs.
2. In the IGSS 8.0 Start menu, double-click the ODBC Server icon.
3. Double-click it once more to make it visible.



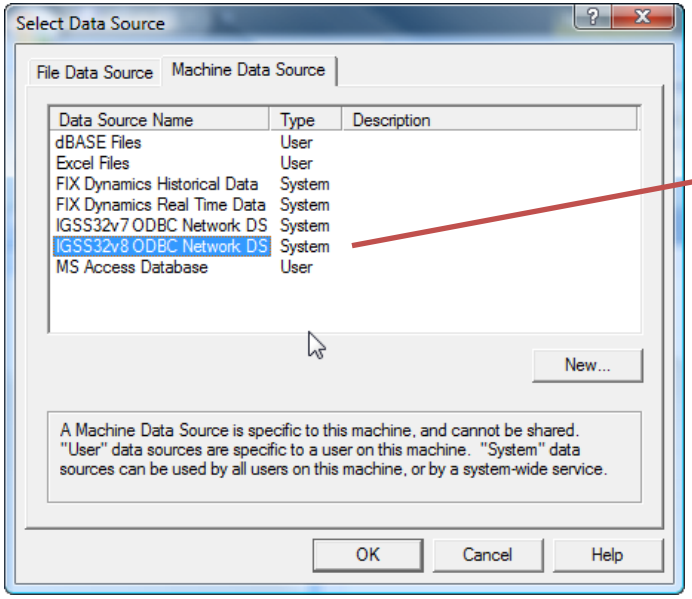
Configure a project with ODBC (2)

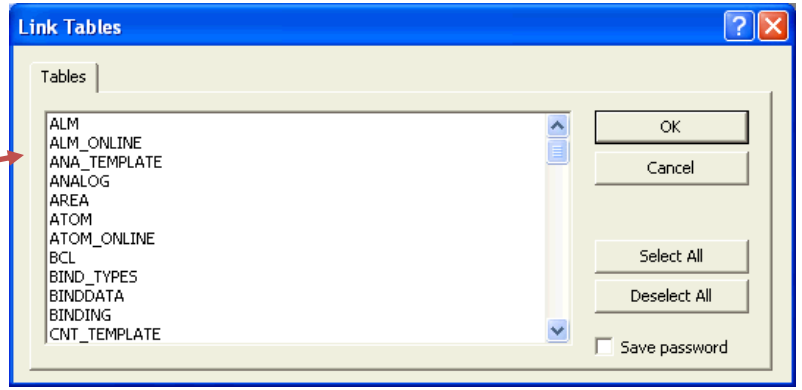
1. Start **Microsoft Access 2007**.
2. Create a new database and name it **Demo_ODBC.accdb** and place it in the project folder.
3. On the **External Data** tab, select **More** and choose **ODBC Database**.
4. Choose **Link to the data ….**



Configure a project with ODBC (3)

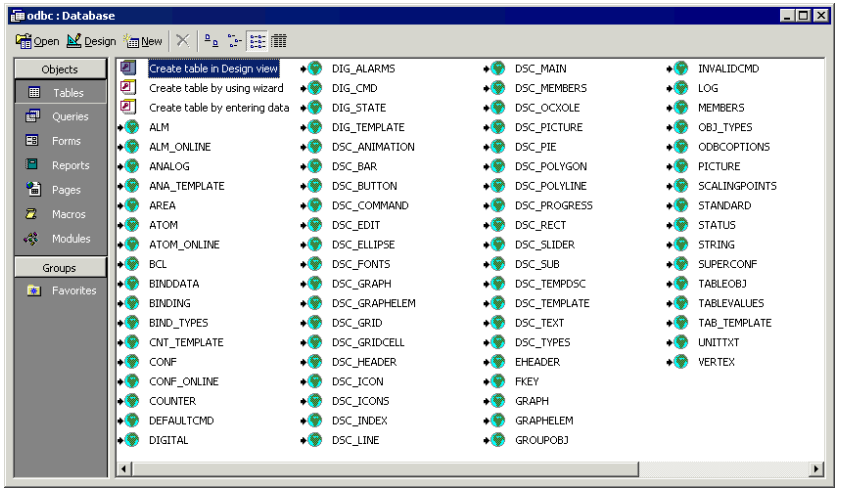
1. In the Select Data Source dialog box, select the Machine Data Source tab.
2. Select the IGSS ODBC Server data source, **IGSS32v8 ODBC Network DS** and click OK.
3. All the IGSS ODBC tables appear in the **Link Tables** dialog box.
4. Click **Select All** and click **OK**.





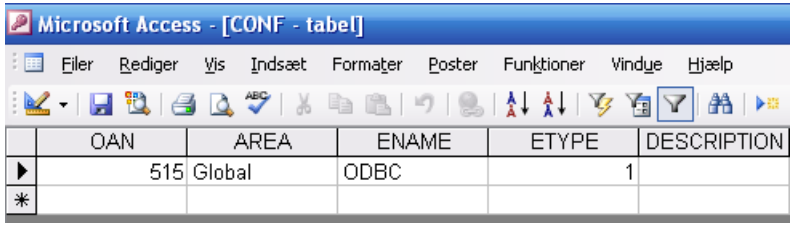
Configure a project with ODBC (4)

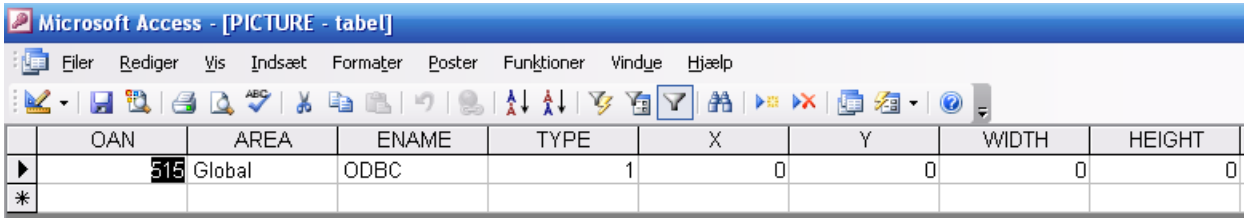
1. Explain the categories of the ODBC tables shown.
   - General tables (CONF, ATOM, SUPERCONF, etc.)
   - Type-specific tables (ANALOG, ANA_TEMPLATE, etc.)
   - Process data tables (ALM, LOG, BCL)
   - Online data tables (ALM_ONLINE, ATOM_ONLINE, CONF_ONLINE)
   - Descriptor Tables (DSC_xxx)



Create a new diagram

1. Open the table named **CONF** to create the new diagram.
2. Click new record and enter **Training** as area, **ODBC** as diagram name, **1** as type.
3. Close the **CONF** table.
4. Open the table named **Picture**.
5. Find the diagram we just created, ODBC.
6. Explain that the properties here are exactly the same as in the **Diagram Properties** dialog box.





Create a new digital object

Creating a new object is even easier, because you can specify all properties from the SUPERCONF table.

1. Open the SUPERCONF table.
2. Find the object p1.
3. Select the entire record and copy.
4. Click the **New Record** button and paste.
5. Modify the properties as follows:
   - Name = p8
   - Node = 1
6. Open the ATOM table and change the PLC address.
7. Close the database.

Insert the new objects in Definition

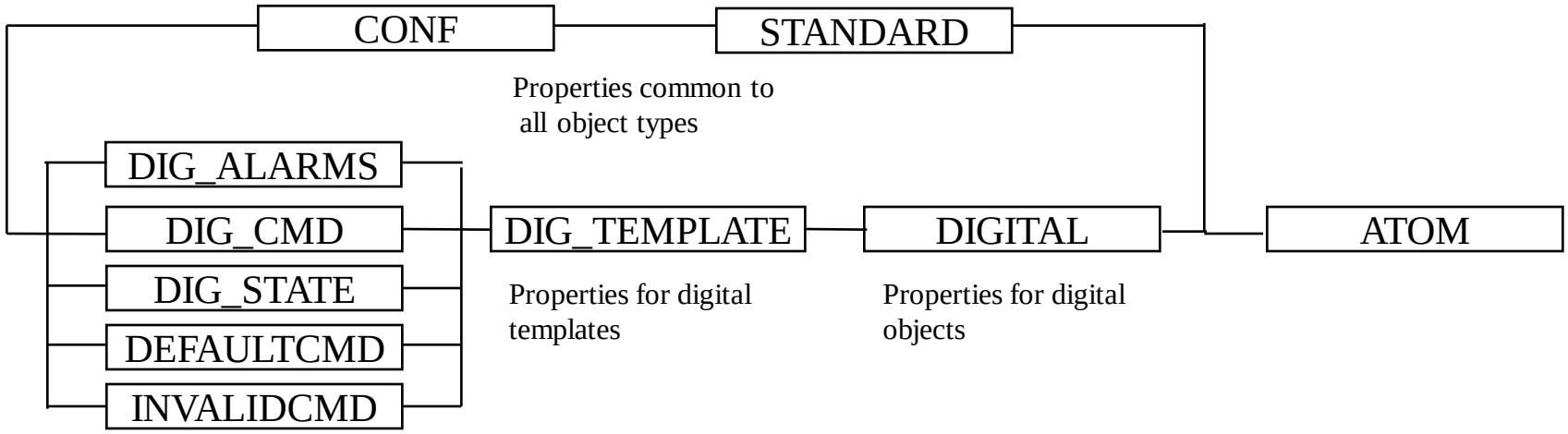
1. Click **Shutdown** in the **IGSS ODBC Server** window.
2. Open the **Definition** module.
3. Go to the **Training** area.
4. Find the new diagram called **ODBC** and open it.
5. Let's insert the new pump called **p1**.
6. Select **Objects > Blowers, Compressors, Pumps**.
7. Select a pump symbol and explain that in the real world you have to change the PLC address.
8. Click **OK** and the new object is there.

END OF STORY

ODBC database structure

1. It is not possible to view the table relationships directly in a database.
2. Therefore, 7T has made a database table overview in the presentation that you can download and in the Interface Help file.

Digital objects



Further information

Please refer to the **ODBC** section of the **Interface Help** file found in the **IGSS 8.0** Start menu.

